Unsupervised Domain Adaption of Hyperspectral Images Based on Paring Domain Discrimination

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***Abstract*— Unsupervised domain adaptation (UDA) reduces domain shifts between distributions to enable model generalization to new scenarios. Adversarial domain adaptation (DA) is an effective approach that extracts domain-invariant features through adversarial learning, but such methods often neglect the influence of category differences on domain discrimination. To solve this problem, we construct a new unsupervised domain adaption hyperspectral image (HSI) classification method. The proposed method consists of two modules, namely, the pairing domain discrimination learning module and the multilevel mutual information maximization module. We propose to construct the sample pair as the input of the domain discriminator. We introduce a new label to the sample pair according to the labels of the two samples and use the relationship between samples to reduce the impact of sample category differences on domain discrimination. When extracting the shared features of the two domains, it will inevitably cause the loss of task-related information. This information is retained by maximizing the proposed multilevel mutual information. The experimental results on different datasets show the effectiveness of our method.**

***Index Terms*— Hyperspectral images (HSIs), mutual information, pairing domain discrimination, unsupervised domain adaptation (UDA).**

I. Introduction

LAND use and land cover (LULC) mapping is playing an important role in land resource management, urban planning, precision agriculture, and environmental protection [1]. Hyperspectral images (HSIs) contain hundreds of continuous spectral bands with narrow bandwidths. Thus, HSIs classification has brought new opportunities for LULC mapping [2]. Meanwhile, with the development of convolutional neural network (CNN), the performance of HSI classification has improved significantly [3], [4], [5]. However, these methods rely on a large number of labeled samples, and the annotation of the dataset is expensive, laborious, and time-consuming. In reality, it is often encountered that the target dataset lacks or even does not have labels, in which case the traditional methods of HSI classification do not achieve promising results. To solve this problem, the other datasets, which are similar to the target dataset, are introduced into the training process to assist the classification tasks, where the other datasets are called the source domain and the target datasets are called the target domain. However, due to the distribution differences between the source domain and the target domain, the classification model directly trained by the source domain does not work well in the target domain.

In order to solve the problem of distribution difference between the source domain and the target domain, the concept of domain adaptation (DA) is introduced. The methods of domain adaption use the knowledge learned from the source domain to improve the performance of the model in the target domain [6], [7], [8], [9]. In reality, the labels of the target domain are always not known in prior, and it is not convenient and efficient to label the target samples. Thus, unsupervised domain adaption has attracted much attention [10], [11]. Due to the influence of environmental factors, such as weather, terrain, and time, hyperspectral data collected from various scenes may exhibit significant distributional differences, rendering traditional classification methods unsuitable. In response, unsupervised hyperspectral DA methods have garnered widespread use for classification tasks in hyperspectral data.

Most methods are based on the idea of the generated adversarial network. The domain discriminator tries to distinguish whether the samples belong to the source domain or the target domain, and the feature extraction module tries to confuse the two domain samples. Through the adversarial training process, domain-independent features can be extracted. Although these methods achieve the distribution adaptation to a certain extent, they all classify a single sample by the domain discriminator, and the classification results are easily affected by domain-independent features, because the input samples not only have domain differences but also belong to different categories. When the domain discriminator classifies the features extracted from the input samples in the domain, it confuses different domain samples. Thus, it may also confuse samples of different categories, resulting in a decline in the classification performance of the model in the target domain. In addition, the unsupervised DA (UDA)

Received 10 April 2024; revised 26 June 2024 and 13 October 2024; accepted 15 November 2024. Date of publication 22 November 2024; date of current version 3 December 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62471233, Grant U23B2006, and Grant 62071233; in part by Jiangsu Provincial Innovation Support Program under Grant BZ2023046; in part by Jiangsu Provincial Key Research and Development Program under Grant BE2022065-2; and in part by the Open Foundations of Jiangsu Province Engineering Research Center of Airborne Detecting and Intelligent Perceptive Technology under Grant JSECF2023-03. *(Corresponding author: Yang Xu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3502782

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assumes that the source domain and the target domain are in different marginal distributions and the same conditional distribution. In the common feature subspace, the model achieves the marginal distribution alignment of the source domain and the target domain. However, it cannot guarantee that the features can maintain the original conditional distribution in the common subspace. That is, the transformed features may be misclassified. Therefore, only relying on extracting domain-invariant features, the classification performance in the target domain cannot be guaranteed. Thus, it is necessary to retain as much information about classification tasks as possible while aligning features.

In order to mitigate the impact of the above problems, the samples are paired as the input of the domain discriminator. According to whether the two samples belong to the same domain, new labels are added to the generated pairing samples. Through the domain discrimination of sample pairs, on the one hand, the relationship between two samples is introduced in the domain discrimination process. For the samples in the same domain, the domain discriminator can better extract the domain-specific features shared by the samples and reduce the impact of domain-invariant features on feature alignment, which are extracted from the samples belong to different categories. On the other hand, the training samples are expanded to improve the stability of the model. In DA, it is insufficient to just extract the domain-invariant features of the source domain and the target domain, and more information related to classification tasks should be retained. It is unrealistic to retain relevant information directly, since the training process does not involve the information. Therefore, task-related information is indirectly retained by maximizing the mutual information between the input and extracted features.

By considering the above problems, we propose a new unsupervised cross-domain HSI classification network. Given two datasets belonging to the same category space, we apply the feature extractor to generate the corresponding representation and then use the domain discriminator to align the data distribution of the source domain and the target domain. Before inputting into the domain discriminator, we pair the samples in pairs, the source domain with the source domain, the target domain with the target domain, and the source domain with the target domain. New labels are attached according to whether samples are from the same domain. At the same time, in order to retain as much relevant information related to the classification task as possible, we introduce new constraints on the extracted features to maximize the multilevel mutual information between the output features of each network layer. The highlights of the proposed method are threefold as follows.

1) We propose a new unsupervised cross-domain HSI classification network. A pairing domain discrimination is developed, which reduces the domain difference between the source domain and the target domain by distinguishing the domain of sample pairs.
2) To mitigate the impact of category inconsistencies among samples from different domains, we employ the classifier to calculate the probability that two samples share the same class and use it as a weight for the sample pairs during training.
3) In order to retain as much information about classified tasks as possible, multilevel mutual information maximization is introduced into the method, which can prevent the model overfitting.

The rest of this article are organized as follows. Section II introduces the latest development of UDA of HSIs. Section III introduces the specific implementation of the proposed unsupervised cross-domain HSI classification network. Section IV gives the experimental results of this method on four different cross-domain-adaptive scenarios. The conclusions are given in Section V.

II. Related Work

A. HSI Classification

The main task of HSI classification is to assign a label to each pixel vector of the cube according to the spectral or spatial characteristics of the image. At present, a lot of works have been done in HSI classification. The methods of HSI classification based on CNN can be divided into three categories: CNN based on spectral information, CNN based on spatial information, and CNN based on spectral–spatial information.

CNN based on spectral information uses 1-D spectral information of images to achieve image classification. Yu et al. [12] proposed a deep convolution network to solve the problem of overfitting, and the 1-D convolution kernel is used to obtain better generalization ability. Gao et al. [13] converted a 1-D spectral vector into a 2-D characteristic matrix, the influence of correlation between bands on classification is weakened, and cascade composite layers are constructed to achieve feature reuse. Charmisha et al. [14] analyzed the impact of dimensionality reduction on vectorized convolution neural network (VCNN) and used dynamic mode decomposition (DMD) to convert 3-D data into 2-D data.

CNN based on spatial information extracts spatial information by reducing the dimension of the HSI image in the spectral domain. Li et al. [15] used PCA to obtain the spatial information and input these data into the convolutional network to achieve classification. Ding et al. [16] proposed to cut patches from an HSI as the input of the 2D-CNN classification model. Chen et al. [17] combined a CNN and Gabor filtering to classify hyperspectral data to alleviate the problem of overfitting.

CNN based on spectral–spatial information combines spatial information and spectral information to achieve the spectral–spatial pixel-level classification of HSIs. Zhong et al. [18] proposed an end-to-end spectral–spatial residual network (SSRN), which uses spectral and spatial residual blocks to learn rich spectral–spatial information. In order to reflect the changes in spatial background information in different spectral blocks, Li et al. [19] proposed an adaptive spectral–spatial feature learning network to learn the weight of each pixel in the spectral block. Limited labeled training samples will lead to the problem of overfitting. Zhang et al. [20] designed an

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end-to-end 3-D lightweight CNN (3D LWNet) to achieve higher classification performance at lower computing cost. Zhang et al. [21] designed a dual-channel CNN framework, in which 1-D CNN extracts spectral features and 1-D CNN extracts spatial features. Feng et al. [22] proposed the multicomplementary generative adversarial networks (GANs) with contrastive learning to extract complementary multiscale spatial–spectral features.

B. Unsupervised DA

Due to various factors in computer vision, such as lighting, background, and weather conditions, there are distribution differences between the source domain and the target domain. The traditional classification models assume that the training data and test data are in the same distribution; therefore, the model trained on the source domain data cannot perform well on the target domain. To solve this problem, the concepts of DA were proposed. DA assumes that the source domain and target domain are in different marginal distributions, but the source domain and the target domain belong to the same category space. At present, the methods of UDA are roughly divided into three categories: sample adaptation, feature adaptation, and model adaptation.

The sample adaptation selects samples similar to the target domain in the source domain and attaches higher weights to these samples, and it obtains a data distribution closer to the target domain by resampling. Yan et al. [23] proposed the weighted maximum mean difference (WMMD) model; to solve the problem of inconsistent class prior distribution between the source domain and the target domain, they introduced the class-specific auxiliary weight into the model. Long et al. [24] believed that feature matching and instance reweighting are equally important when there is a large difference between the two domains. Based on this assumption, a transfer joint matching (TJM) method was proposed to achieve cross-domain image recognition.

The feature adaptation realizes the migration between the source domain and the target domain by obtaining the common feature representation between different domains. Gopalan et al. [25] selected a group of subspaces according to the geodesic line between the source domain and the target domain and found a suitable intermediate representation to achieve low-dimensional subspace migration. Zhang et al. [26] proposed an unsupervised domain-adaptive method to simultaneously reduce geometric offset and distribution offset, which is called joint geometric and statistical alignment (JGSA). The source domain and target domain data are projected into a low-dimensional subspace through two projections. Das and Lee [27] regarded the source domain and the target domain as hypergraphs and used the similarity between the graphs to match the two domains. Ni et al. [28] obtained a set of subspaces to connect the source domain and the target domain through dictionary learning and introduced a metric method of the difference between the two domains to select the most appropriate domain. Long et al. [29] believed that label alignment is also required when aligning features and proposed a conditional adversarial DA network (CDAN), which realizes DA through joint distribution alignment of the features and the categories.

Based on the similarity between classification models in different domains, the model adaptation modifies the loss function in the source domain and transfers the trained model in the source domain to the target domain. Duan et al. [30] designed a cross-domain kernel learning framework, called domain transfer multiple kernel learning (DTMKL). This method realizes distribution adaptation between the source domain and the target domain by minimizing structural loss function and distribution difference. Cao et al. [31] proposed a new distribution matching machine (DMM), which uses manifold and MMD to extract invariant features. Gholami et al. [32] constructed an unsupervised probabilistic latent variable model (PUnDA), which realized domain alignment by learning potential projection spaces and classifiers of different domains.

C. Cross-Domain HSI Classification

Based on the assumption that the training data and the target data are in the same distribution, the traditional hyperspectral classification methods rely on a large number of labeled data. However, on the one hand, labeling HSIs are time-consuming and labor-intensive; on the other hand, due to sensor, time, background, weather, and other factors, there is a spectral shift between different hyperspectral datasets [33].

In order to solve the above problems, transfer learning is applied to HSI classification, that is, cross-domain HSI classification. By transferring the knowledge learned from the source domain to the target domain, the model can perform well on the unlabeled target domain. Rajan et al. [34] introduced a binary hierarchical classifier (BHC) into the model to transfer knowledge from the source domain to the target domain. Sun et al. [35] introduced domain transfer multikernel learning to solve the problem of distribution difference between the source domain and the target domain. Persello and Bruzzone [36] minimized the distance between two domains in the reproducing kernel Hilbert space. When the difference between the two domains is small, the methods based on machine learning can achieve promising results in cross-domain HSI classification. However, when the difference between the two domains is large, these methods cannot perform well.

In order to solve this problem, scholars turn their perspective to in deep learning. Deng et al. [37] proposed a cross-domain hyperspectral classification model based on depth metric learning, which uses adversarial methods to realize DA and uses the similarity between sample pairs to classify. Liu et al. [38] constructed a classwise distribution adaption (CDA) network, which computes the maximum mean difference for each class to achieve class antagonism adaptation. Feng et al. [39] designed a split-level capsule network to extract spatial–spectral features and a sampler for category imbalance. Fang et al. [40] proposed a domain-adaptive method based on confident learning (CLDA), which uses confident learning to obtain high-confidence target samples as training samples to improve the classification ability of the network. TSTnet [8] utilizes the graph to obtain topological

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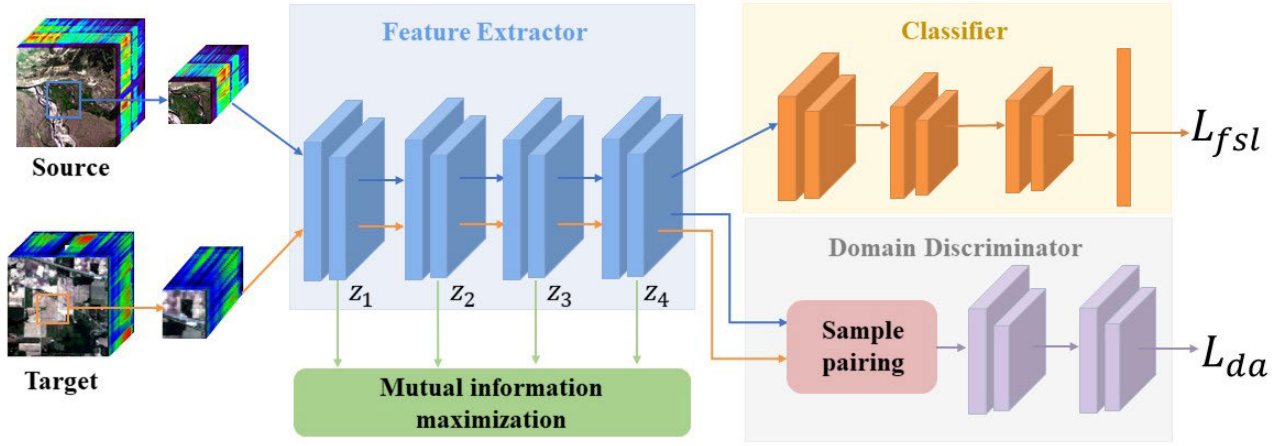


Fig. 1. Architecture of the proposed method. The model consists of four modules, feature extractor, classifier, domain discriminator, and mutual information maximization module, which are divided by different colors.

relationships between different feature classes. The model achieves distribution alignment by combining graph-optimal transport and MMD to improve model robustness. Considering the separability of the data, Li et al. [41] utilized supervised contrastive learning to enhance the performance of the model. Based on this method, they further proposed a cross-modal alignment framework for HSIs and text prototypes [42], which uses text semantics to guide the model to obtain a more accurate representation of image prototypes.

Distributional differences are caused by a variety of factors, and it is unrealistic to obtain all possible distributions of the target domain in practical applications. The model needs to achieve good generalization performance even under unknown target domains. Zhang et al. [43] proposed a single-source domain extension network (SDEnet), which introduces a generator utilizing a semantic encoder and morph encoder to generate an extended domain through spatial and spectral randomization. To utilize linguistic modal knowledge, they also proposed a language-aware domain generalization network (LDGnet) [44], which utilizes a dual-stream architecture with image and text encoders to extract both coarse-grained and fine-grained linguistic features. It aligns visual and linguistic information through supervised contrastive learning in a shared semantic space.

III. Pairing Domain Discrimination HSI Classification

A. Problem Formulation

Given a labeled dataset and an unlabeled dataset, the purpose of unsupervised cross-domain HSI classification is to train a classifier that performs well in the target domain through labeled source domain data and unlabeled target domain data. In particular, the source domain is defined as $\boldsymbol{D}_S = \{(\boldsymbol{x}_i^S, \boldsymbol{y}_i^S)\}_{i=1}^{N_S}$, where $\boldsymbol{x}_i^S$ and $\mathbf{y}_i^S$, respectively, represent the images and the labels of the source domain, and the target domain is defined as $\mathbf{D}_T = \{(\mathbf{x}_i^T)\}_{i=1}^{N_T}$, where $\mathbf{x}_i^T$ represents the images of the target domain. Unlike the source domain, the target domain does not contain label information. Due to the distribution difference between the source domain and the target domain, the classification model trained directly by the source domain data cannot perform well in the target domain.

The traditional cross-domain unsupervised classification methods, based on GANs, typically classify individual samples. However, their classification results are easily affected by domain-invariant features. While confusing the features of the source domain and the target domain, the features of samples, which belong to different categories, are also easy to be confused, leading to the classification performance degradation of the model in the target domain. To solve this problem, we pair the samples according to different domains and use the extracted features of the paired sample as the input of the domain discriminator, and it uses the relationship between samples to reduce the interference of domain-invariant features. When distributions are adapted, the samples of two domains are mapped to the common subspace. However, in the process of extracting shared features between the two domains, some task-related information for the classification task may be lost, resulting in misclassified samples. In order to retain the classification task-related information as much as possible, we use a multilevel mutual information maximization module to maximize the mutual information between the input and the output. The proposed module is shown in Fig. 1.

Given the source domain image $\mathbf{x}_s$ and the target domain image $\mathbf{x}_t$, we first use the feature extractor to obtain the sample representation, where the output of each layer can be expressed as $\mathbf{z}_i$, $(i = 1, 2, 3, 4)$. Next, we pair the sample representations from each domain and feed them into a domain discriminator. At the same time, we apply adversarial learning between the feature extractor and the domain discriminator to narrow the distribution difference. Finally, the output of the feature extractor is input into the multilevel mutual information maximization module to retain the classification task information.

In the following, we describe the domain discrimination module and multilevel mutual information maximization module in detail.

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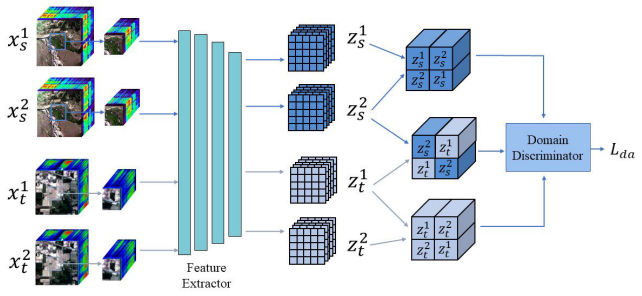


Fig. 2. Process of pairing domain discrimination.

B. Pairing Domain Discrimination Module

Currently, there are many adversarial methods for achieving distribution alignment between the source and target domains. However, most of these methods rely on domain recognition at the level of individual samples. This approach is prone to confusion between different categories of samples and may inadvertently extract domain-invariant features that lead to misclassification of target domain samples. To solve this problem, we propose a pairing domain discrimination module.

We generate a cube pair by assembling the feature maps from the same domain or different domains into a pair, which is expressed as follows:

$$\mathbf{P}_{ij} = \text{Pair}\left(\mathbf{F}(\mathbf{x}_i), \mathbf{F}\left(\mathbf{x}_j\right)\right) \tag{1}$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ represent the training samples, $\mathbf{F}(\bullet)$ represents the feature extractor, and Pair$(\bullet)$ connects the two feature maps to form a cube pair. Determine the label of the cube pair according to whether samples $\mathbf{x}_i$ and x_j belong to the source domain or the target domain

$$\mathbf{Y}_{ij} = \begin{cases} 0, & \text{if } \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ belong to the source domain} \\ 1, & \text{if } \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ belong to the target domain} \\ 2, & \text{if } \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ belong to different domains} \end{cases} \tag{2}$$

where $\mathbf{Y}_{ij}$ represents the label of the cube pair $\mathbf{P}_{ij}$.

The process of pairing domain learning is shown in Fig. 2. We use a fully CNN as the domain discriminator, which classifies the cube pairs and calculates the domain discrimination loss based on the prediction results. However, since the two samples used to construct a cube pair are randomly selected, it cannot be guaranteed that they belong to the same category. This can sometimes result in domain recognition of cube pairs containing different categories of samples, which may distort the feature distribution of the target domain. To address this issue, we use a classifier to predict the probability that two samples belong to the same category and weigh each cube pair accordingly. This helps to mitigate the impact of different categories and improve the accuracy of our pairing domain discrimination approach. Then, the domain discrimination loss can be expressed as follows:

$$L_{\text{da}}\left(\mathbf{P}_{\text{ij}}, \mathbf{Y}_{\text{ij}}\right) = E_{\mathbf{p}_{\text{ij}} \sim \mathbf{P}_{\text{ij}}} \mathbf{w}_{\text{ij}} * \sum_{c=1}^{3} \mathbf{y}_{\text{ij}} \log\left(\mathbf{D}_{\text{da}}\left(\mathbf{p}_{\text{ij}}\right)\right) \tag{3}$$

$$\mathbf{w}_{\text{ij}} = \frac{1}{C} \sum_{c=1}^{C} \mathbf{D}_{\text{cls}}(\mathbf{F}(\mathbf{x}_i)) * \mathbf{D}_{\text{cls}}\left(\mathbf{F}\left(\mathbf{x}_j\right)\right) \tag{4}$$

where $\mathbf{P}_{\text{ij}}$ represents the label set corresponding to the sample set $\mathbf{Y}_{\text{ij}}$; $\mathbf{p}_{\text{ij}}$ and $\mathbf{y}_{\text{ij}}$ represent the cube pair and corresponding label, respectively; $\mathbf{w}_{\text{ij}}$ represents the weight of the cube pair; $\mathbf{D}_{\text{da}}(\bullet)$ and $\mathbf{D}_{\text{cls}}(\bullet)$ represent the domain discriminator and the classifier, respectively; C represents the number of the categories; c represents the specific class; and $\mathbf{F}$ represents the feature extractor.

C. Multilevel Mutual Information Maximization Module

One of the most common methods in unsupervised cross-domain HSI classification tasks is to identify domain-invariant features. This involves training a network that can extract shared features from different domains and then using these domain-invariant features to train a classifier. While this approach can transfer knowledge learned from the source domain to the target domain, it does not consider the potential information loss caused by feature mapping. In order to find the optimal common subspace and extract the shared features of both domains, the model removes as much domain-specific information as possible, which includes information related to classification tasks. Unfortunately, this includes information that is relevant to classification tasks. As a result, the loss of this information can lead to a decline in the overall classification performance of the model. Therefore, how to retain the information related to classification tasks in representational learning is very important.

During model training, it is unrealistic to expect that we can directly retain classification task-related information from the representation of data without explicitly leveraging this information during training. Information loss is an inherent challenge in feature extraction processes and becomes more pronounced with deeper network layers. As a result, the input layer generally contains the most information related to classification tasks, while subsequent layers may progressively lose some of this information as they extract more abstract features. Based on this assumption, we can change the goal from retaining the relevant information of the classification task to ensuring the maximization of mutual information between the extracted representation and the input. We introduce a multilevel mutual information maximization module into the model. This module uses the Donsker–Varadhan (DV) [45] representation based on KL divergence to obtain the lower bound of mutual information, which is proposed by the mutual information neural estimation (MINE) [46]. The formula is as follows:

$$\begin{aligned} I(\mathbf{X}; \mathbf{Y}) &:= D_{\text{KL}}(J \| M) \geq \\ \hat{I}_{\omega}^{(\text{DV})}(\mathbf{X}; \mathbf{Y}) &:= E_J\left[\mathbf{D}_{\text{MI}}(\mathbf{x}, \mathbf{y})\right] - \log E_M\left[e^{\mathbf{D}_{\text{MI}}(\mathbf{x}, \mathbf{y})}\right] \end{aligned} \tag{5}$$

where $\mathbf{X}$ and $\mathbf{Y}$ represent the image and encoded representation, respectively; J represents the joint distribution $P_{\mathbf{XY}}$ of $\mathbf{X}$ and $\mathbf{Y}$; M represents the product of the marginal distribution $P_{\mathbf{X}} P_{\mathbf{Y}}$; and $\mathbf{D}_{\text{MI}}$ represents a discriminator with a parameter of ω, which is used to determine whether $\mathbf{X}$ and $\mathbf{Y}$ are representations from the same sample.

According to this formula, the goal of maximizing mutual information can be converted into maximizing the lower bound of mutual information. We can maximize the lower bound of

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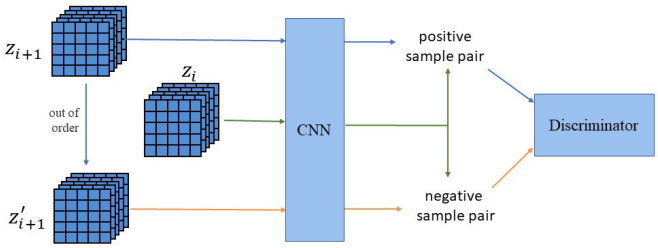


Fig. 3. Architecture of the multilevel mutual information maximization module. $\mathbf{z}_i$ and $\mathbf{z}_{i+1}$ represent the outputs of different network layers from the same sample, and $\mathbf{z}'_{i+1}$ is obtained by disturbing the sequence order of $\mathbf{z}_{i+1}$. In this way, $\mathbf{z}_i$ and $\mathbf{z}'_{i+1}$ are paired to obtain a batch of representation pairs from different samples, that is, negative sample pairs. Positive representation pairs are obtained by pairing $\mathbf{z}_i$ and $\mathbf{z}_{i+1}$. The discriminator distinguishes whether the input representation pairs come from the same sample, so as to maximize the mutual information between the representations of the same sample.

mutual information through adversarial training. As shown in Fig. 3, we choose two representations $\mathbf{z}_i$ and $\mathbf{z}_{i+1}$ of the same image to form a positive representation pair. One representation $\mathbf{z}_{i+1}$ is shuffled to obtain $\mathbf{z}'_{i+1}$, $\mathbf{z}_i$ and $\mathbf{z}'_{i+1}$ are formed into a negative representation pair, and the two representation pairs are input into the mutual information discriminator $\mathbf{D}_{\mathrm{MI}}$ for judgment. Finally, the probability of two representations coming from the same sample is obtained. Therefore, the loss of mutual information maximization is

$$\begin{aligned}\mathrm{MI}(\mathbf{z}_i, \mathbf{z}_{i+1}) = {} & \mathrm{SoftPlus}(\mathbf{D}_{\mathrm{MI}}(\mathbf{z}_i, \mathbf{z}_{i+1})) \\ & - \mathrm{SoftPlus}\left(\mathbf{D}_{\mathrm{MI}}\left(\mathbf{z}_i, \mathbf{z}'_{i+1}\right)\right)\end{aligned} \tag{6}$$

where SoftPlus is an activation function $\mathrm{SoftPlus}(x) = \log(1 + e^x)$, and $\mathbf{z}_i$ and $\mathbf{z}_{i+1}$ are two representations of the same sample.

In the final mutual information calculation, mutual information between different feature levels is summed together. Thus, we name the mutual information as multilevel mutual information. Because the shallow network tends to extract the domain features of the samples and the deep network tends to extract the features related to the classification tasks of the samples, we assign different weights when calculating the mutual information between layers. The loss of the multilevel mutual information maximization module is

$$L_{\mathrm{MI}} = 0.25 * \mathrm{MI}(\mathbf{z}_1, \mathbf{z}_2) + 0.5 * \mathrm{MI}(\mathbf{z}_2, \mathbf{z}_3) + \mathrm{MI}(\mathbf{z}_3, \mathbf{z}_4). \tag{7}$$

D. Network Training

The proposed method consists of two modules, pairing domain discrimination and multilevel mutual information maximization. The network includes a classifier, a domain discriminator, and a mutual information discriminator. The loss of classified tasks is

$$L_{\mathrm{cls}}(\mathbf{X}_S, \mathbf{Y}_S) = \frac{1}{N_S} \sum_{i=1}^{N_S} L_{\mathrm{CE}}\left(\mathbf{D}_{\mathrm{cls}}\left(\mathbf{F}\left(\mathbf{x}_i^s\right)\right), \mathbf{y}_i^s\right) \tag{8}$$

where $\mathbf{x}_i^s$ and $\mathbf{y}_i^s$ are the samples of the source domain, $\mathbf{D}_{\mathrm{cls}}$ represents the classifier that distinguishes which category the sample belongs to, and L_{CE} represents the loss of cross entropy.

In addition, we add a regularization constraint to the target domain to make its prediction probability peak

$$L_{\mathrm{reg}}(\mathbf{X}_T) = -\frac{1}{n_t} \sum_{x_i^t \in \mathbf{X}_T} \sum_{c=1}^{C} p_i^c \log p_i^c \tag{9}$$

where p_i^c indicates the probability that sample $\mathbf{x}_i^t$ belongs to class c.

The training process of DA is divided into three parts. First, the overall network is updated, and its loss function can be written as follows:

$$L = L_{\mathrm{cls}}(\mathbf{X}_S, \mathbf{Y}_S) + \alpha L_{\mathrm{da}} + \beta L_{\mathrm{MI}} + \gamma L_{\mathrm{reg}}(\mathbf{X}_T) \tag{10}$$

where α, β, and γ are hyperparameters.

Second, the feature extraction network is updated using samples from both the source and target domains. The feature extraction network attempts to confuse samples from both domains, and the loss can be written as follows:

$$L = L_{\mathrm{reg}} + L_{\mathrm{da}}. \tag{11}$$

Finally, updating the domain discriminator to distinguish between source and target domain samples and the loss can be written as follows:

$$L = L_{\mathrm{da}}. \tag{12}$$

TABLE I
SAMPLE DISTRIBUTION OF THE HOUSTON DATASET

Class	Name	Houston 2013	Houston 2018
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365

TABLE II
SAMPLE DISTRIBUTION OF THE PAVIA DATASET

Class	Name	UP	PC
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584

IV. EXPERIMENTS

A. Empirical Setup

1) Datasets: To illustrate the effectiveness of our method, we conducted experiments in four cross-domain-adaptive scenarios, Houston 2013→Houston 2018, UP→PC, PC→UP, and Dioni→Loukia.

The Houston datasets consist of data collected by different sensors in different years, namely, Houston 2013 and Houston 2018. Houston 2013 has a data size of 349×1905 pixels with 144 spectral bands, while Houston 2018 has a data size of

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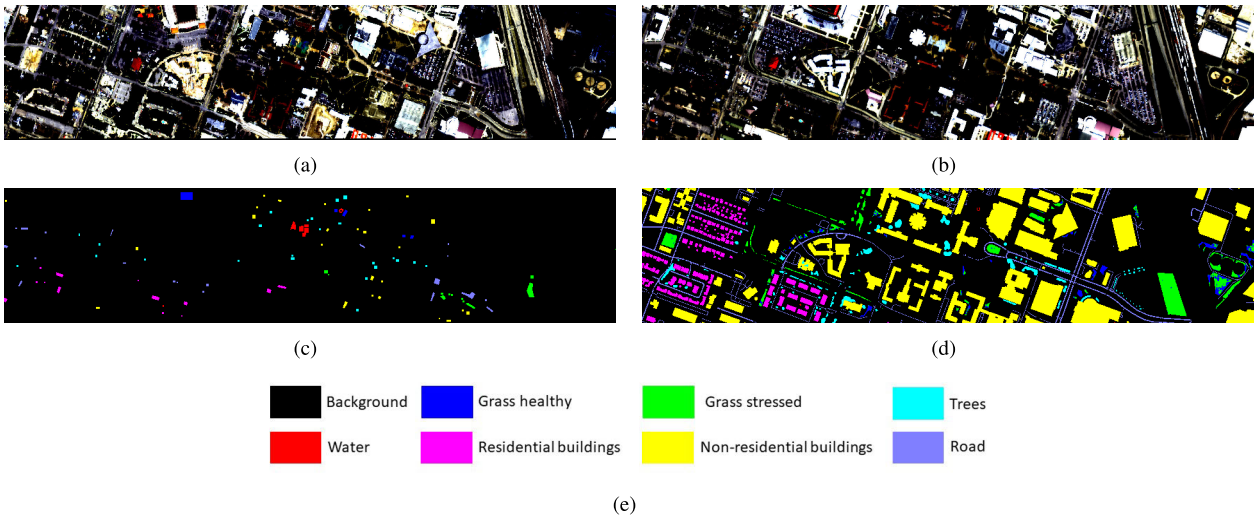


Fig. 4. Houston. (a) Houston 2013 false-color image. (b) Houston 2018 false-color image. (c) Houston 2013 ground-truth map. (d) Houston 2018 ground-truth map.

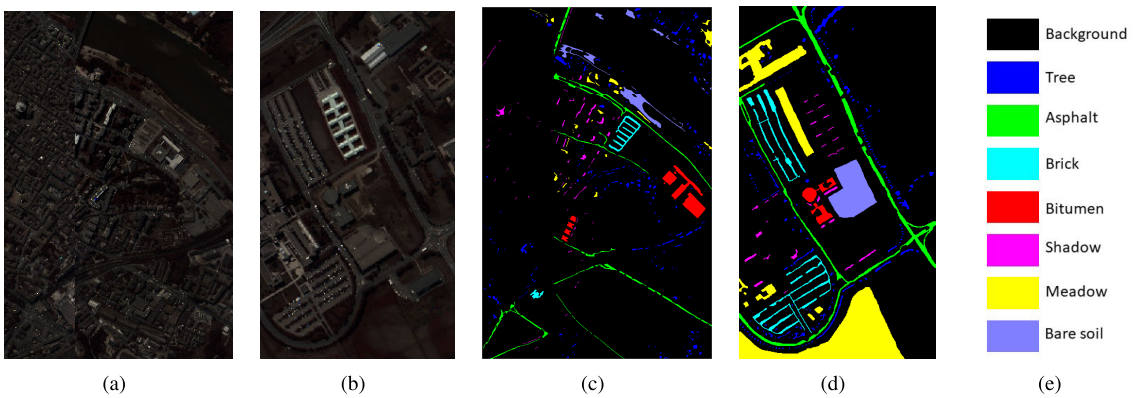


Fig. 5. Pavia. (a) PC false-color image. (b) UP false-color image. (c) PC ground-truth map. (d) UP ground-truth map.

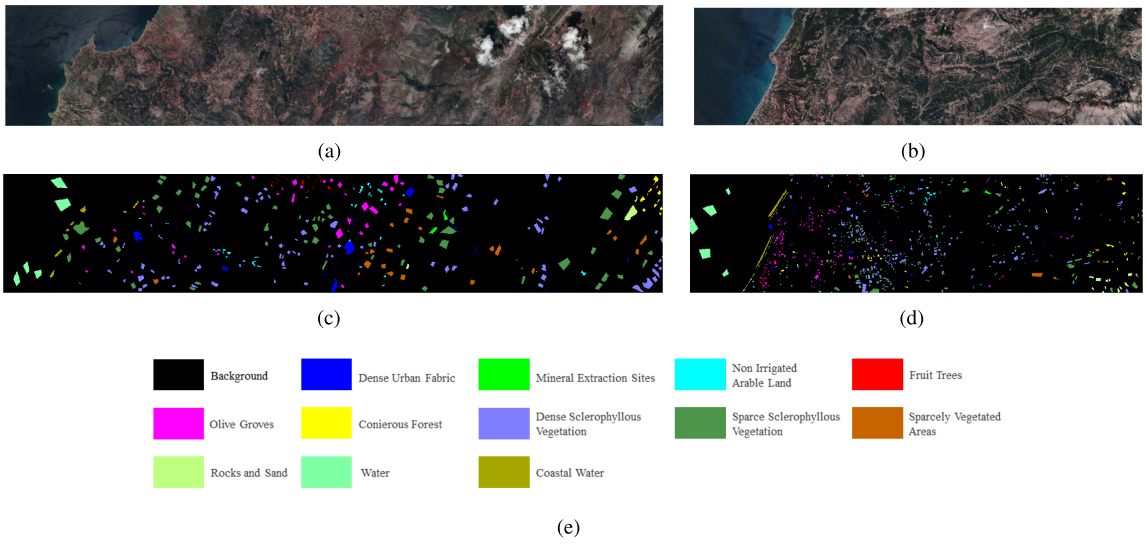


Fig. 6. HyRANK. (a) Dioni false-color image. (b) Loukia false-color image. (c) Dioni ground-truth map. (d) Loukia ground-truth map.

209 × 955 pixels and only 49 spectral bands. For this study, we designate Houston 2013 as the source domain and Houston 2018 as the target domain. To ensure consistency between the source and target domains, we selected a 209 × 955 area in Houston 2013 and retained only the 48 spectral bands that were common to both datasets. The filtered dataset contains seven shared classes, which are detailed in Table I. Fig. 4 displays the pseudo-RGB images and ground-truth maps.

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TABLE III
SAMPLE DISTRIBUTION OF THE HYRANK DATASET

Class	Name	Dioni	Loukia
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Conierous Forest	361	422
7	Dense Sclerophyllous Vegetation	5035	2996
8	Sparce Sclerophyllous Vegetation	6374	2361
9	Sparcely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421

TABLE IV
STRUCTURE OF OUR NETWORK

	Layer Name	Input Dimension	Output Dimension
Feature Extractor	3x3 Conv+BN+ReLU	bz x d x 5 x 5	bz x 64 x 5 x 5
	1x1 Conv+BN+ReLU	bz x 64 x 5 x 5	bz x 64 x 5 x 5
	1x1 Conv+BN+ReLU	bz x 64 x 5 x 5	bz x 64 x 5 x 5
	1x1 Conv+BN+ReLU	bz x 64 x 5 x 5	bz x 64 x 5 x 5
	AvgPool	bz x 64 x 5 x 5	bz x 64 x 1 x 1
Classifier	1x1 Conv+BN+ReLU	bz x 64 x 1 x 1	bz x 32 x 1 x 1
	1x1 Conv+BN+ReLU	bz x 32 x 1 x 1	bz x 32 x 1 x 1
	FC	bz x 32	bz x C
Discriminator	2x2 Conv+BN+ReLU	bz x 64 x 4 x 4	bz x 32 x 1 x 1
	1x1 Conv+BN+ReLU	bz x 32 x 1 x 1	bz x 16 x 1 x 1
	FC	bz x 16	bz x 3

The Pavia University (UP) is the hyperspectral data collected at the University of Pavia. It has a data size of 610×340 with 103 wavebands. The Pavia Center (PC) is the data collected in the center of Pavia City. It has a data size of 1096×715 with 102 wavebands. In order to ensure an equal number of bands in the two datasets, we removed the last band from the UP dataset. We experimented with the two datasets as the source domain and the target domain, respectively. The two datasets contain seven shared classes, which are detailed in Table II. Fig. 5 displays the pseudo-RGB images and ground-truth maps.

The Dioni is the hyperspectral data collected by Hyperion. The size of this dataset is 250×1376, and it contains 176 wavebands. The Loukia is also collected by Hyperion. Its size is 249×945, including 176 wavebands. We take Dioni as the source domain and Loukia as the target domain. The two datasets contain 12 shared classes, which are detailed in Table III. Fig. 6 shows their pseudo-RGB images and ground-truth maps.

2) Metrics: We use three indicators to evaluate the model: overall accuracy, average category accuracy, and kappa coefficient.

B. Implementation Details

The structure of our network is shown in Table IV, where bz, d, and C represent the batch size, number of image channels, and number of categories, respectively. We choose five comparison methods to illustrate the effectiveness of our method, including DANN [47], CDAN [29], MDD [48], ED-DMM-UDA [37], CLDA [40], FDDAN [49], and CACL [50]. In all experiments, we selected 180 samples from each category of the source domain to ensure the fairness of the experiment. We use SGD of 0.9 momentum and set the initial learning rate to 0.001. For all datasets, we set the number of epochs to 200. Due to the randomness of sample selection, each experiment is conducted ten times to take the average value as the final result.

TABLE V
CLASSIFICATION RESULTS (%) OF THE HOUSTON DATASET

Class	DANN	MDD	CDAN	ED-DMM-UDA	CLDA	FDDAN	CACL	OURS
1	47.82	72.95	80.50	68.14	64.97	67.89	**89.06**	65.53
2	**88.76**	86.85	84.12	85.45	88.04	68.95	81.28	86.84
3	57.74	69.99	72.21	57.86	71.73	**79.52**	66.09	60.31
4	36.36	81.82	**100**	79.55	93.16	**100**	95.45	60.45
5	**98.20**	91.40	80.75	97.77	96.13	79.66	94.69	93.63
6	51.35	55.71	46.14	56.62	61.71	54.05	66.51	**70.76**
7	75.40	**85.04**	81.01	75.97	77.71	75.41	32.36	56.52
OA	62.61±2.10	66.86±3.75	59.53±2.88	66.09±1.83	70.12±2.43	62.25±3.24	67.18±2.19	**72.15±1.12**
AA	65.09±3.54	77.68±3.60	77.82±0.69	74.48±1.66	**79.07±0.38**	75.07±2.45	75.06±1.75	70.58±2.87
Kappa	49.33±1.98	54.36±4.36	47.19±2.50	53.10±2.18	**57.75±2.49**	48.75±3.04	52.23±1.33	57.71±1.55

TABLE VI
CLASSIFICATION RESULTS (%) OF THE PC DATASET

Class	DANN	MDD	CDAN	ED-DMM-UDA	CLDA	FDDAN	CACL	OURS
1	88.73	86.57	91.17	**98.21**	97.27	95.43	93.67	94.30
2	87.63	83.85	95.12	96.63	98.54	98.93	97.42	**99.16**
3	75.20	85.68	71.87	72.40	**93.70**	85.18	73.00	85.22
4	56.74	50.65	71.73	67.94	82.07	85.01	85.15	**86.03**
5	99.99	99.73	99.97	**100**	**100**	**100**	**100**	99.99
6	60.99	75.81	82.02	25.46	68.97	**92.62**	88.25	90.67
7	67.24	72.19	79.38	83.92	81.06	91.84	93.81	**93.84**
OA	76.67±4.92	76.94±3.96	85.13±5.01	82.50±0.50	92.80±0.67	93.14±1.06	91.62±1.36	**93.34±0.76**
AA	76.65±4.37	79.23±2.56	84.47±5.09	77.80±0.56	91.83±1.09	92.72±2.18	89.91±1.61	**92.74±0.73**
Kappa	71.99±5.75	72.56±4.57	82.07±6.12	78.73±0.54	91.32±0.81	91.75±1.25	90.19±0.65	**91.98±0.90**

TABLE VII
CLASSIFICATION RESULTS (%) OF THE UP DATASET

Class	DANN	MDD	CDAN	ED-DMM-UDA	CLDA	FDDAN	CACL	OURS
1	90.64	83.35	77.84	92.07	89.97	86.36	62.55	**92.90**
2	55.42	64.35	**86.58**	72.98	81.61	59.19	83.17	58.89
3	92.75	71.34	91.93	97.06	**98.56**	48.70	93.36	91.84
4	70.78	79.89	14.14	82.82	**87.48**	78.12	38.04	86.57
5	99.86	98.70	95.25	99.28	**100**	99.05	95.30	98.30
6	49.75	55.51	46.13	48.97	43.84	53.45	**64.17**	39.11
7	66.04	53.89	84.39	71.88	83.90	84.51	58.81	**84.57**
OA	61.92±7.28	62.31±2.67	64.70±2.06	66.16±5.62	66.87±3.77	62.44±4.00	69.16±2.74	**69.41±4.59**
AA	75.03±1.87	72.43±3.88	70.89±1.21	80.72±2.50	**83.62±4.59**	72.77±2.85	70.77±4.15	78.88±2.28
Kappa	52.46±7.18	51.95±3.02	55.31±1.95	57.44±5.25	**58.85±3.84**	52.56±4.12	58.75±2.04	51.56±4.76

C. Experimental Results

For DANN, it is necessary to standardize the samples before training the network, aligning the distributions of samples from the source domain and the target domain to follow a standard normal distribution. In order to make the model fully trained, the number of training epochs is set to 1000. For ED-DMM-UDA, its training process consists of two stages, training on the source domain and then on the target domain. To achieve a well-clustered training set, the number of training epochs is set to 200 for the source domain and 20 for the target domain. In order to ensure the fairness of the experiment, the patch size for all datasets is set to 5, and 180 samples of each category in the source domain and all unlabeled samples in the target domain were selected in all experiments.

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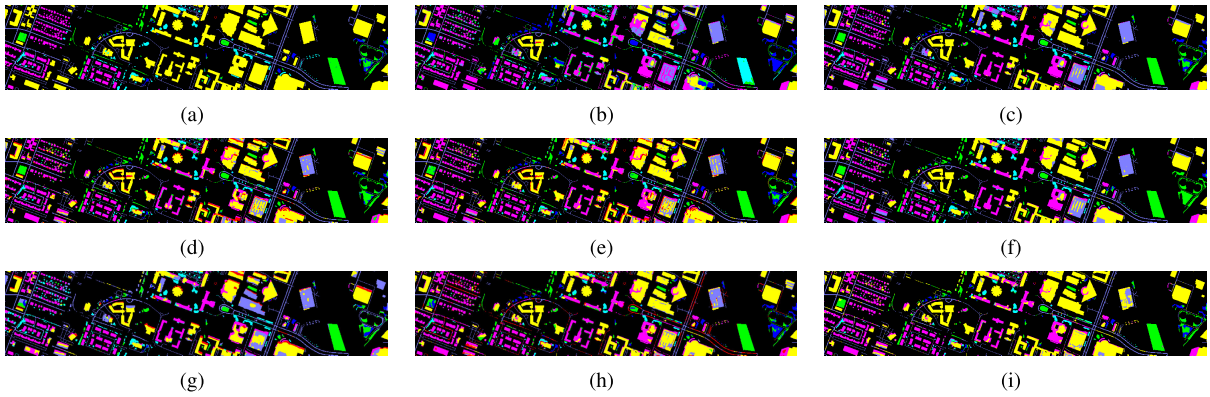


Fig. 7. Houston 2018. (a) Ground-truth map. (b)–(i) Classification results for different methods. (b) DANN. (c) MDD. (d) CDAN. (e) ED-DMM-UDA. (f) CLDA. (g) FDDAN. (h) CACL. (i) Our method.

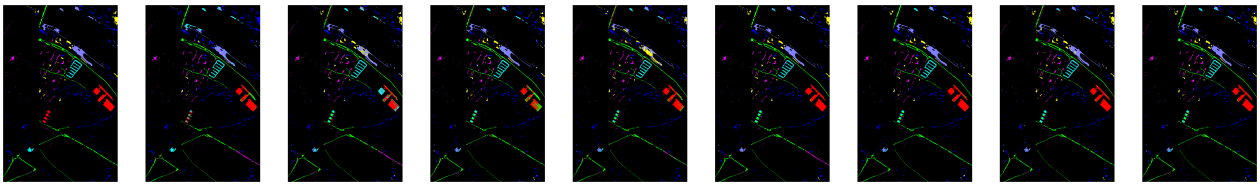


Fig. 8. PC. (a) Ground-truth map. (b)–(i) Classification results for different methods. (b) DANN. (c) MDD. (d) CDAN. (e) ED-DMM-UDA. (f) CLDA. (g) FDDAN. (h) CACL. (i) Our method.

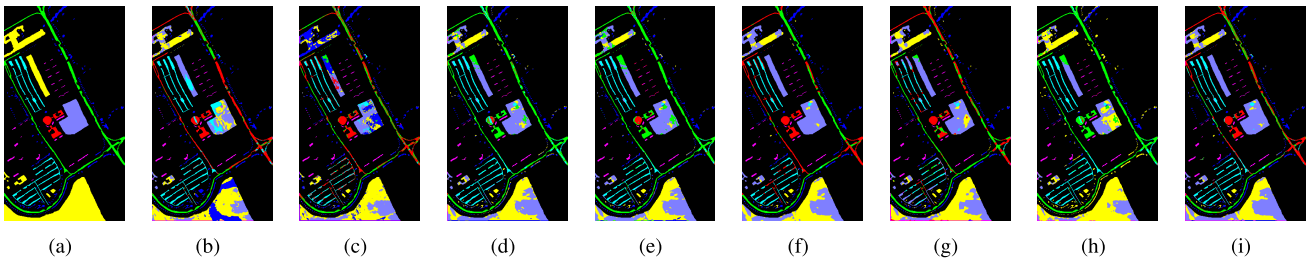


Fig. 9. UP. (a) Ground-truth map. (b)–(i) Classification results for different methods. (b) DANN. (c) MDD. (d) CDAN. (e) ED-DMM-UDA. (f) CLDA. (g) FDDAN. (h) CACL. (i) Our method.

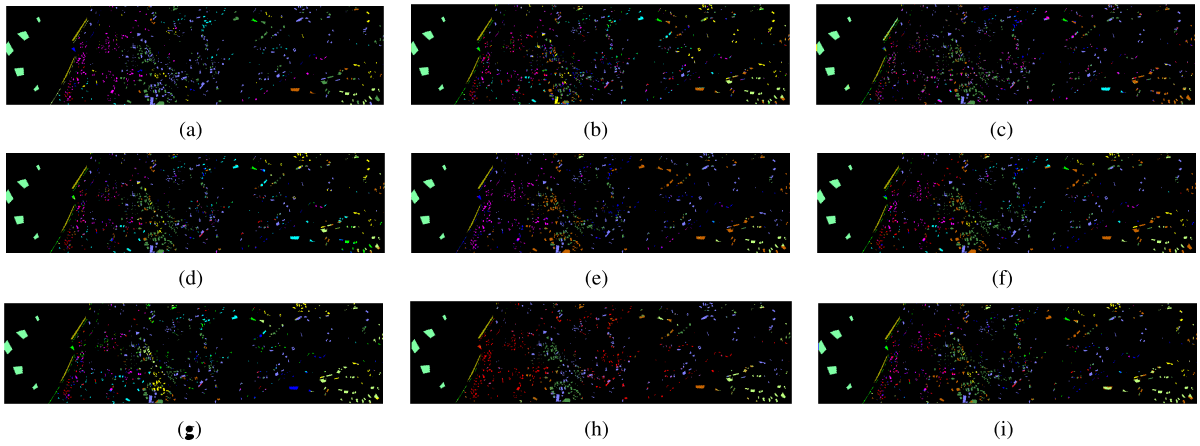


Fig. 10. Loukia. (a) Ground-truth map. (b)–(i) Classification results for different methods. (b) DANN. (c) MDD. (d) CDAN. (e) ED-DMM-UDA. (f) CLDA. (g) FDDAN. (h) CACL. (i) Our method.

Tables V–VIII show the OA, AA, and kappa of different methods on different datasets as well as the classification accuracy of each category. In order to eliminate the impact of random sampling on the experimental results, all experiments were carried out ten times, and the mean of them was taken as the final classification results.

The DANN comprises a classifier and a domain discriminator. The classifier computes the classification loss of the

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TABLE VIII
CLASSIFICATION RESULTS (%) OF THE LOUKIA DATASET

Class	DANN	MDD	CDAN	ED-DMM-UDA	CLDA	FDDAN	CACL	OURS
1	21.36	50.05	27.72	**51.26**	43.01	29.13	18.93	45.92
2	94.44	78.89	98.89	22.04	90.37	**100**	7.41	99.63
3	43.50	**50.68**	47.16	6.55	11.81	38.03	5.04	30.56
4	77.09	64.05	74.81	9.62	74.30	**77.22**	70.88	75.82
5	39.86	40.86	21.44	**65.40**	36.21	40.47	26.29	44.48
6	48.65	29.79	38.53	4.79	26.33	**60.90**	1.31	51.73
7	33.82	38.21	49.42	70.33	58.16	39.59	**78.13**	57.00
8	14.43	18.85	33.90	27.25	39.52	21.26	**39.88**	38.30
9	34.56	21.30	20.73	**90.78**	57.09	13.28	75.57	69.42
10	48.98	17.09	21.28	10.04	29.01	71.74	**74.50**	64.77
11	88.96	91.92	**100**	92.71	99.01	**100**	**100**	98.61
12	96.63	98.15	93.73	**99.86**	99.71	99.76	50.35	99.22
OA	42.14±1.82	43.00±4.16	48.64±2.90	55.90±8.12	54.20±3.38	47.70±4.11	57.61±1.71	**58.38±1.87**
AA	53.52±3.20	49.99±3.49	52.30±4.85	45.89±4.47	55.38±5.11	57.61±3.80	49.70±2.82	**64.62±3.70**
Kappa	35.64±1.87	36.04±3.89	40.74±3.37	48.63±7.99	46.71±3.80	41.43±4.25	45.69±3.90	**51.95±2.11**

TABLE IX
CLASSIFICATION RESULTS (%) OF FOUR DATASET. P INDICATES PAIRING DOMAIN DISCRIMINATION. M INDICATES MULTILEVEL MUTUAL INFORMATION MAXIMIZATION. L INDICATES REGULARIZATION LOSS

P	M	L	H13→H18	UP→PC	PC→UP	Dioni→Loukia
✓			69.14	88.46	65.94	56.21
	✓		68.14	88.13	65.79	56.64
		✓	67.86	86.60	64.51	55.92
✓	✓		71.84	92.91	68.73	58.11
	✓	✓	70.25	92.47	67.85	58.29
✓		✓	71.67	93.07	68.94	58.16
✓	✓	✓	**72.15**	**93.34**	**69.41**	**58.38**

source domain. Based on the idea of adversarial learning, the domain discriminator tries to distinguish whether the samples belong to the source domain or the target domain. Through feature extraction network, DANN confuses the two domains' samples to achieve distribution alignment. Margin disparity discrepancy (MDD) implements DA by measuring the difference between two domains. It introduces marginal differences into the DA method based on adversarial learning. The condition domain adversarial network (CDAN) uses the discriminant information predicted by the classifier to assist the adversarial DA and further realizes the DA by introducing the condition information on the basis of the original adversarial network. Because of the introduction of category information, CDAN is better than DANN in some datasets, but its category information may lead to errors in the prediction process and lead to a decline in the classification performance of the model.

The feature embedding model based on depth metric learning (ED-DMM-UDA) carries out similarity learning on samples and learns the representation information of categories by distinguishing whether two samples belong to the same category. The method realizes DA between two scenarios by aligning samples in source domain and target domain based on contrastive methods. The CLDA first labels the target domain with false labels and then uses confident learning to select samples with high confidence to add to the training set. In order to reduce the distribution difference between the source domain and the target domain, this method introduces two classifiers into the network and extracts the shared features of the two domains by minimizing the prediction difference between the two classifiers. Different from CDAN, this method alleviates the impact of wrong samples on the classification model, and it can also be found from the table that the results of CLDA on four datasets are better than CDAN. In order to enhance the attention to category features, FDDAN introduces two disentangles to extract domain-invariant features specific to each class. Based on the consistency between the learned class prototypes and the predicted results, CACL proposes inter-/intra-domain contrast learning to achieve fine-grained feature alignment. Our method considers the influence of category information on domain recognition in the process of domain discrimination and introduces multilevel mutual information maximization to retain the information related to classification tasks. Compared with other methods, our method has the highest classification accuracy. Figs. 7–10 show the classification prediction graph of different methods. By comparison, our method can correctly classify more samples, and the prediction diagram is closer to the truth diagram.

D. Ablation Experiment

To illustrate the effectiveness of the module, we show the performance of pairing domain discrimination, multilevel mutual information maximization, and the regularization loss (P, M, and L in Table IX) on four datasets. From Table IX, we can find that the classification accuracy of the method introducing DA into the model is much higher than that of the method without any modules. The modules further improve the classification performance of the model in the target domain.

The left-hand side of Fig. 11 shows the distribution of the original target samples, while the right-hand side shows the visualization results of the extracted features. From Fig. 11, we can find that the features extracted by the feature extractor have separability, indicating that our method can improve the generalization performance of the model.

Figs. 12 and 13 demonstrate the effect of the three hyperparameters α, β, and γ as well as the number of training samples, on the accuracy of the model. Fig. 12 illustrates the trend of the accuracy of our method on the four datasets when α and β are combined differently. From the figure, it can be seen that a value of 0.5 for α and 0.1 for β achieves relatively good performance on all four datasets. Fig. 13(a) shows the variation of the accuracy of our method on the four datasets when the value of the hyperparameter γ is varied, from which it can be seen that the PC and UP datasets achieve high accuracy when γ is taken as 0.05, and the highest accuracy is achieved on the Houston and HYRANK datasets when γ is taken as 0.1. Fig. 13(b) demonstrates the effect of the number of training samples on the model accuracy; with the increase in the number of training samples, the model classification accuracy shows an overall upward trend.

In Table X, we compared the training and testing time of different methods. Our method requires more training time over DANN, MDD, CDAN, and CACL, comparable training time with CLDA, and less training time over ED-DMM-UDA

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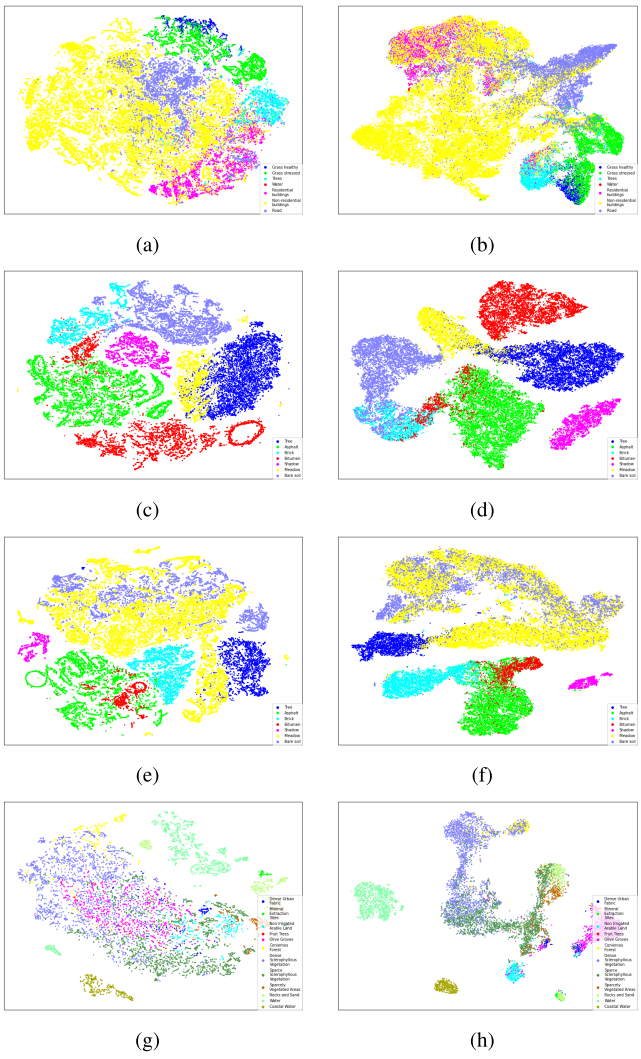


Fig. 11. T-SNE visualization. The left images are original samples from target domains, and the right images are extracted features from target domains. (a) and (b) Houston 2013→Houston 2018. (c) and (d) UP→PC. (e) and (f) PC→UP. (g) and (h) Dioni→Loukia.

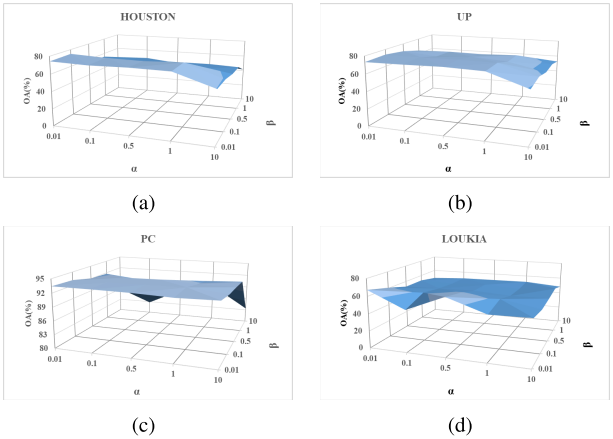


Fig. 12. Sensitivity analysis of parameters α and β in four datasets. (a) Houston 2013→Houston 2018. (b) PC→UP. (c) UP→PC. (d) Dioni→Loukia.

and FDDAN. Still, the test time of the proposed method is the longest. But, the overall test time of all methods is at

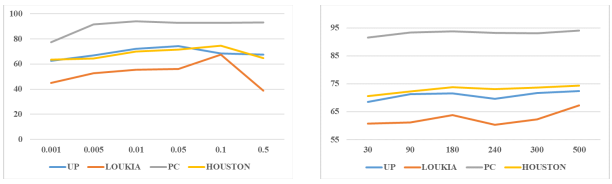


Fig. 13. Sensitivity analysis of parameters in four datasets. (a) Parameter γ. (b) Number of training samples.

TABLE X
TIME COMPARISON (S) OF EACH DATASET BETWEEN DIFFERENT METHODS

		DANN	MDD	CDAN	ED-DMM-UDA	CLDA	FDDAN	CACL	OURS
Houston	Training time	6.65	13.06	13.92	532.18	147.06	190.19	1564	327.95
	Testing time	4.9	2.44	4.36	4.21	9.47	1.57	3.89	19.21
PC	Training time	7.23	14.41	12.81	130.66	192.26	188.80	1409	328.98
	Testing time	10.14	12.64	9.68	4.35	6.27	1.31	1.99	13.10
UP	Training time	41.4	14.37	12.77	544.19	191.86	189.30	1475	340.08
	Testing time	4.02	6.58	3.88	3.97	6.15	1.32	1.98	13.41
Loukia	Training time	46.70	15.09	11.69	243.73	145.17	351.92	3032	465.39
	Testing time	5.63	6.19	5.53	2.73	1.81	0.39	1.55	2.42

the same scale. It can be concluded that the proposed method achieves the highest classification accuracy with a few extra time computation.

V. CONCLUSION

In this article, we propose an unsupervised cross-domain hyperspectral classification network. The network extracts invariant features of source domain and target domain based on adversarial learning and introduces the relationship between sample pairs to reduce the impact of category differences on domain discrimination. In order to retain the information related to classification tasks as much as possible, we introduce the multilevel mutual information maximization module in the network. The experimental results on different datasets show that our method performs well in unsupervised cross-domain hyperspectral classification.

REFERENCES

[1] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[2] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[3] M. Ahmad, A. M. Khan, R. Hussain, S. Protasov, F. Chow, and A. M. Khattak, "Unsupervised geometrical feature learning from hyperspectral data," in *Proc. IEEE Symp. Ser. Comput. Intell. (SSCI)*, Dec. 2016, pp. 1–6.

[4] S. Liu, Y. Zheng, Q. Du, A. Samat, X. Tong, and M. Dalponte, "A novel feature fusion approach for VHR remote sensing image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 464–473, 2021.

[5] S. Wan, C. Gong, P. Zhong, S. Pan, G. Li, and J. Yang, "Hyperspectral image classification with context-aware dynamic graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 597–612, Jan. 2021.

[6] J. Zhang, Z. Ding, W. Li, and P. Ogunbona, "Importance weighted adversarial nets for partial domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8156–8164.

[7] P. P. Busto and J. Gall, "Open set domain adaptation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 754–763.

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[8] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023.

[9] W. Li, L. Duan, D. Xu, and I. W. Tsang, "Learning with augmented features for supervised and semi-supervised heterogeneous domain adaptation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 6, pp. 1134–1148, Jun. 2014.

[10] K. Saito, K. Watanabe, Y. Ushiku, and T. Harada, "Maximum classifier discrepancy for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3723–3732.

[11] J. Dong, Y. Cong, G. Sun, Z. Fang, and Z. Ding, "Where and how to transfer: Knowledge aggregation-induced transferability perception for unsupervised domain adaptation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1664–1681, Mar. 2021.

[12] S. Yu, S. Jia, and C. Xu, "Convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, Jan. 2017.

[13] H. Gao, Y. Yang, C. Li, H. Zhou, and X. Qu, "Joint alternate small convolution and feature reuse for hyperspectral image classification," *ISPRS Int. J. Geo-Inf.*, vol. 7, no. 9, p. 349, Aug. 2018.

[14] K. S. Charmisha, V. Sowmya, and K. P. Soman, "Dimensionally reduced features for hyperspectral image classification using deep learning," in *Proc. Int. Commun. Conf., Int. Conf. Commun. Cyber Phys. Eng.*, 2018, pp. 171–179.

[15] J. Li, X. Zhao, Y. Li, Q. Du, B. Xi, and J. Hu, "Classification of hyperspectral imagery using a new fully convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 292–296, Feb. 2018.

[16] C. Ding, Y. Li, Y. Xia, W. Wei, L. Zhang, and Y. Zhang, "Convolutional neural networks based hyperspectral image classification method with adaptive kernels," *Remote Sens.*, vol. 9, no. 6, p. 618, Jun. 2017.

[17] Y. Chen, L. Zhu, P. Ghamisi, X. Jia, G. Li, and L. Tang, "Hyperspectral images classification with Gabor filtering and convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2355–2359, Dec. 2017.

[18] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[19] S. Li, X. Zhu, Y. Liu, and J. Bao, "Adaptive spatial–spectral feature learning for hyperspectral image classification," *IEEE Access*, vol. 7, pp. 61534–61547, 2111.

[20] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[21] H. Zhang, Y. Li, Y. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery using a dual-channel convolutional neural network," *Remote Sens. Lett.*, vol. 8, nos. 4–6, pp. 438–447, May 2017.

[22] J. Feng, Z. Gao, R. Shang, X. Zhang, and L. Jiao, "Multi-complementary generative adversarial networks with contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520018.

[23] H. Yan, Y. Ding, P. Li, Q. Wang, Y. Xu, and W. Zuo, "Mind the class weight bias: Weighted maximum mean discrepancy for unsupervised domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 945–954.

[24] M. Long, J. Wang, G. Ding, J. Sun, and P. S. Yu, "Transfer joint matching for unsupervised domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1410–1417.

[25] R. Gopalan, R. Li, and R. Chellappa, "Domain adaptation for object recognition: An unsupervised approach," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 999–1006.

[26] J. Zhang, W. Li, and P. Ogunbona, "Joint geometrical and statistical alignment for visual domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 5150–5158.

[27] D. Das and C. S. G. Lee, "Unsupervised domain adaptation using regularized hyper-graph matching," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 3758–3762.

[28] J. Ni, Q. Qiu, and R. Chellappa, "Subspace interpolation via dictionary learning for unsupervised domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Portland, OR, USA, Jun. 2013, pp. 692–699.

[29] M. Long, Z. Cao, J. Wang, and M. Jordan, "Conditional adversarial domain adaptation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, Eds., 2018, pp. 1647–1657.

[30] L. Duan, I. W. Tsang, and D. Xu, "Domain transfer multiple kernel learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 3, pp. 465–479, Mar. 2012.

[31] Y. Cao, M. Long, and J. Wang, "Unsupervised domain adaptation with distribution matching machines," in *Proc. 32nd AAAI Conf. Artif. Intell. (AAAI), 30th Innov. Appl. Artif. Intell. (IAAI), 8th AAAI Symp. Educ. Adv. Artif. Intell. (EAAI)*, New Orleans, LA, USA, 2018, pp. 2795–2802.

[32] B. Gholami, O. Rudovic, and V. Pavlovic, "PUnDA: Probabilistic unsupervised domain adaptation for knowledge transfer across visual categories," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 3601–3610.

[33] W. Pan, "A survey of transfer learning for collaborative recommendation with auxiliary data," *Neurocomputing*, vol. 177, pp. 447–453, Feb. 2016.

[34] S. Rajan, J. Ghosh, and M. M. Crawford, "Exploiting class hierarchies for knowledge transfer in hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 11, pp. 3408–3417, Nov. 2006.

[35] Z. Sun, C. Wang, H. Wang, and J. Li, "Learn multiple-kernel SVMs for domain adaptation in hyperspectral data," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 5, pp. 1224–1228, Sep. 2013.

[36] C. Persello and L. Bruzzone, "Kernel-based domain-invariant feature selection in hyperspectral images for transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2615–2626, May 2016.

[37] B. Deng, S. Jia, and D. Shi, "Deep metric learning-based feature embedding for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 1422–1435, Feb. 2019.

[38] Z. Liu, L. Ma, and Q. Du, "Class-wise distribution adaptation for unsupervised classification of hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 508–521, Jan. 2021.

[39] J. Feng et al., "Class-aligned and class-balancing generative domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509617.

[40] Z. Fang et al., "Confident learning-based domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527116.

[41] Z. Li et al., "Supervised contrastive learning-based unsupervised domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524017.

[42] Z. Li et al., "Cross-domain few-shot hyperspectral image classification with cross-modal alignment and supervised contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519319.

[43] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[44] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[45] M. D. Donsker and S. R. S. Varadhan, "Asymptotic evaluation of certain Markov process expectations for large time. IV," *Commun. Pure Appl. Math.*, vol. 36, no. 2, pp. 183–212, Mar. 2010.

[46] I. Belghazi, S. Rajeswar, A. Baratin, R. D. Hjelm, and A. C. Courville, "MINE: Mutual information neural estimation," 2018, *arXiv:1801.04062*.

[47] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2030–2096, May 2016.

[48] Y. Zhang, T. Liu, M. Long, and M. Jordan, "Bridging theory and algorithm for domain adaptation," in *Proc. 36th Int. Conf. Mach. Learn.*, vol. 97, May 2019, pp. 7404–7413.

[49] Z. Xin et al., "Feature disentanglement based domain adaptation network for cross-scene coastal wetland hyperspectral image classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 129, May 2024, Art. no. 103850.

[50] K. Ding, T. Lu, Y. Fang, and S. Li, "Consistency-aware customized learning for cross-scene hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2024, pp. 9080–9083.

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